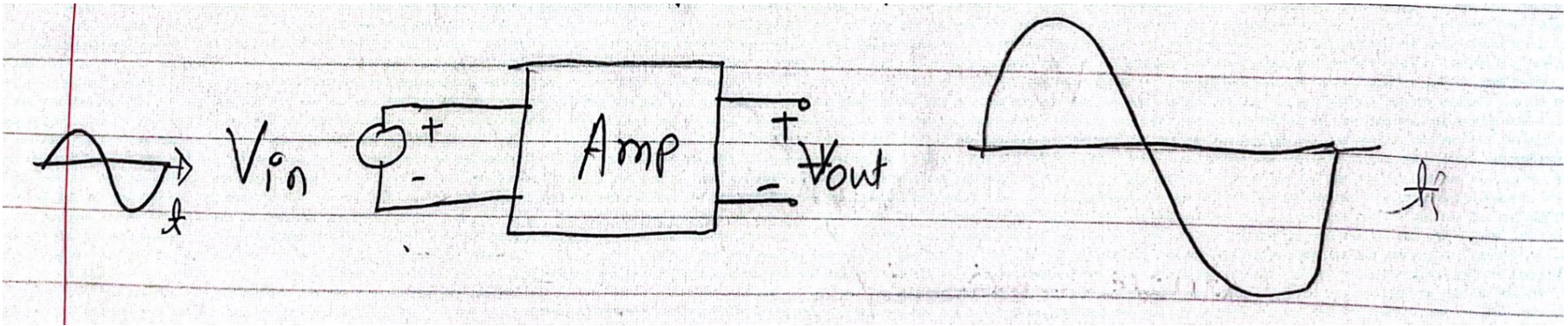


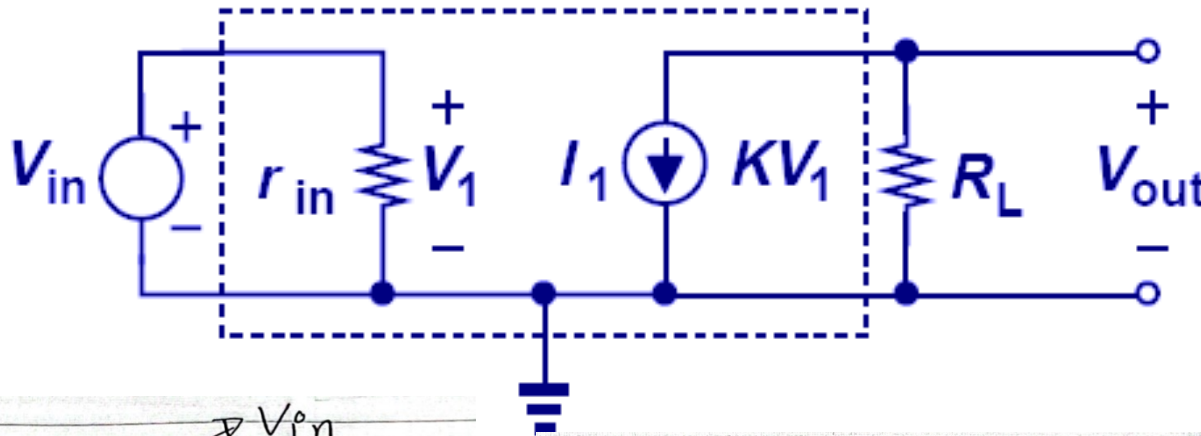
Bipolar Transistor

Voltage-Dependent Current Source



An amplifier is an electronic device that increases the voltage, current, or power of a signal.

Voltage-Dependent Current Source with Input Resistance



$$\frac{V_{out}}{R_L} + KV_1 = 0$$

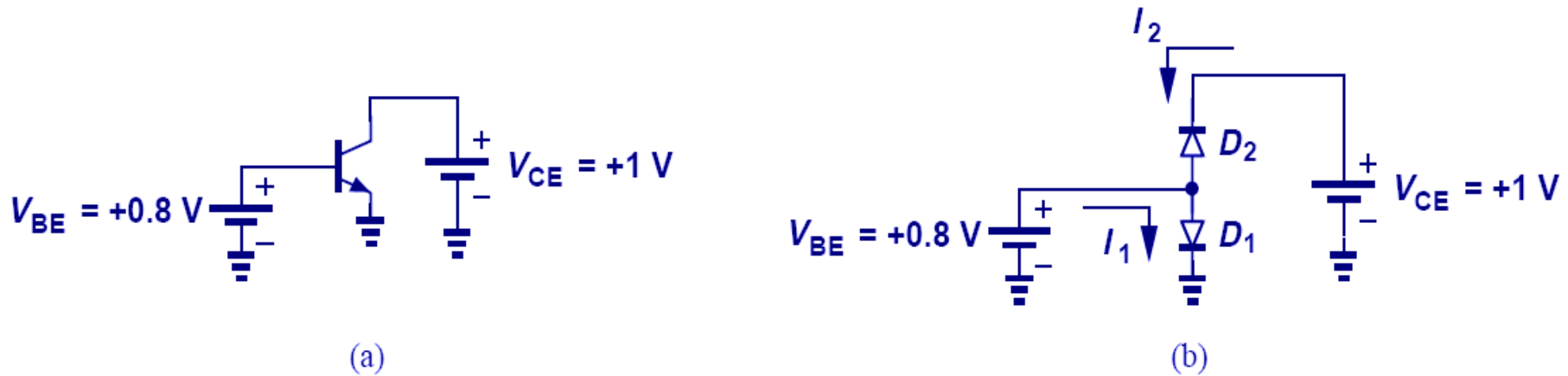
$$\Rightarrow V_{out} = -\underbrace{KR_L}_{-5} V_{in}$$

$$V_{out} = -5 V_{in}$$



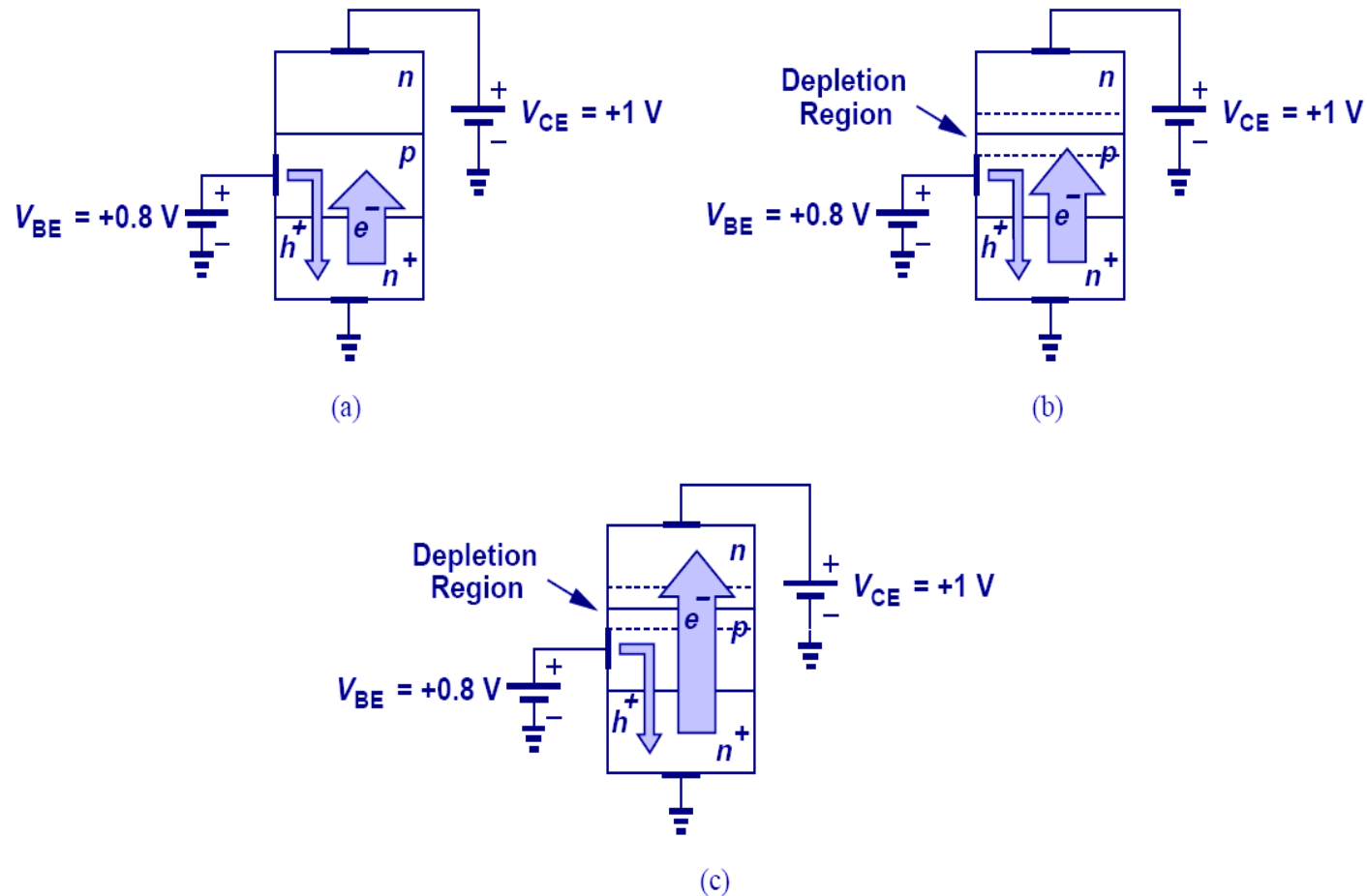
➤ **Regardless of the input resistance, the magnitude of amplification remains unchanged.**

Forward Active Region



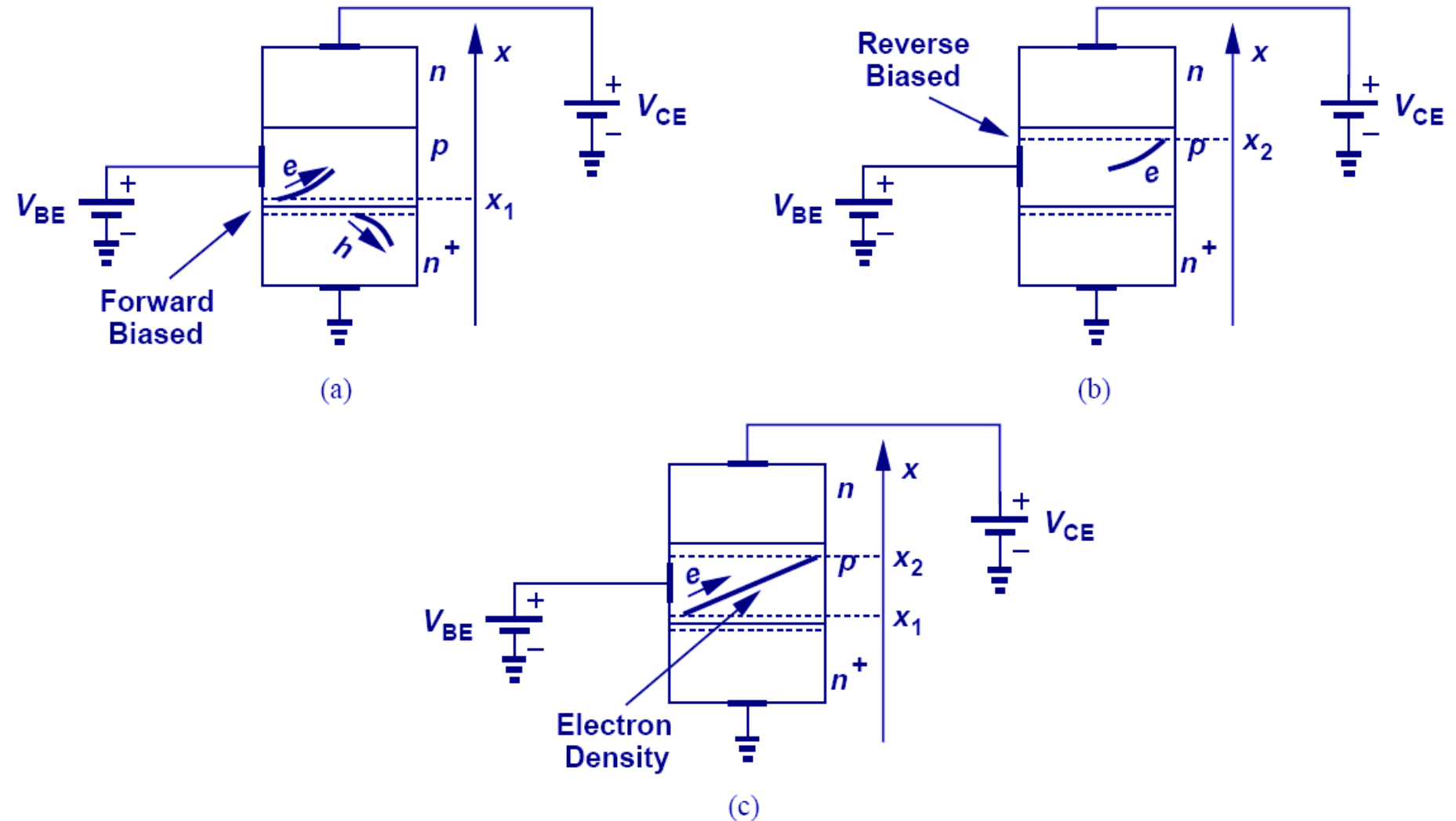
- Forward active region: $V_{BE} > 0$, $V_{BC} < 0$.
- Figure b) presents a wrong way of modeling figure a).

Accurate Bipolar Representation



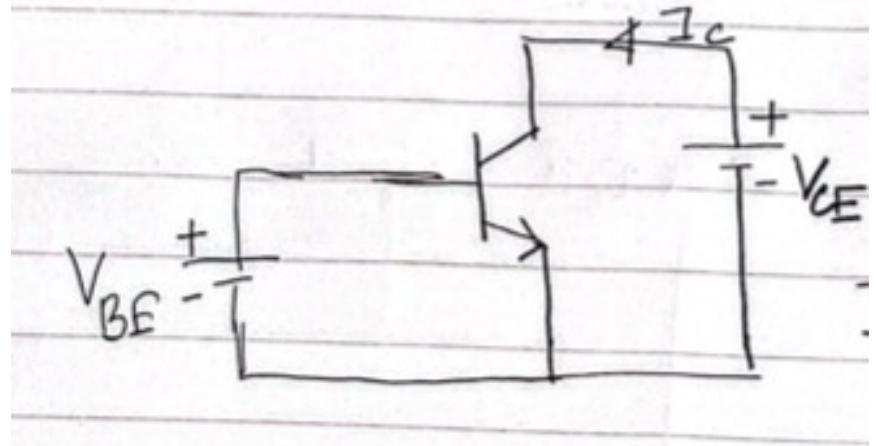
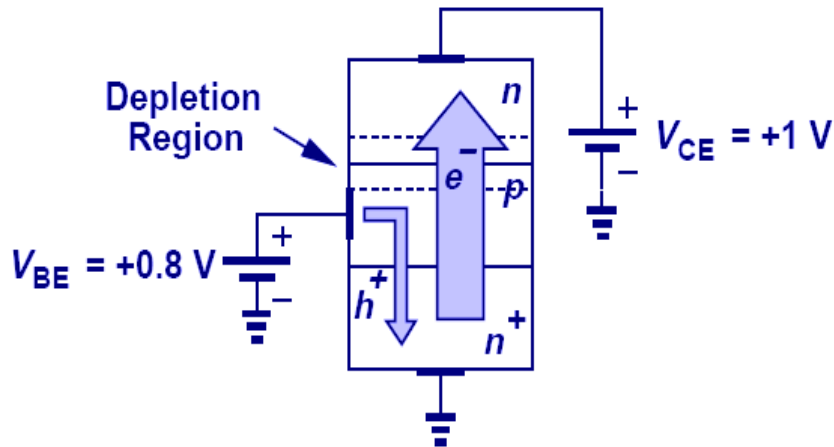
➤ **Collector also carries current due to carrier injection from base.**

Carrier Transport in Base: Drift or Diffusion?



Negligible Drift Current

What happens if V_{BE} increases?



$$I_C = I_S \left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1 \right]$$

$$I_S = \frac{A_E q n_i^2 D_n}{W_B N_B}$$

emitter area

Base width

doping base

Collector Current

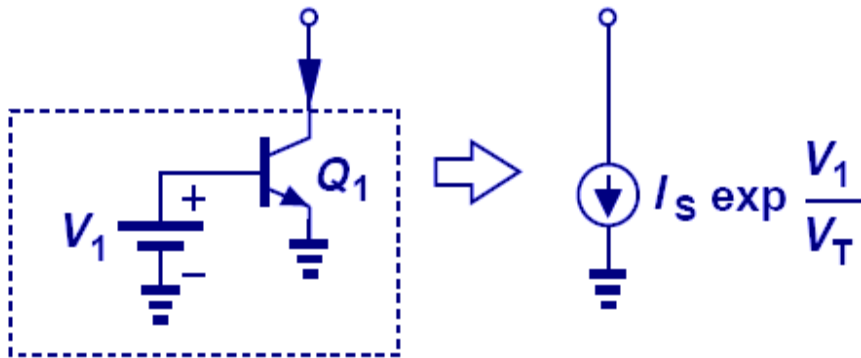
$$I_C = \frac{A_E q D_n n_i^2}{N_E W_B} \left(\exp \frac{V_{BE}}{V_T} - 1 \right)$$

$$I_C = I_S \exp \frac{V_{BE}}{V_T}$$

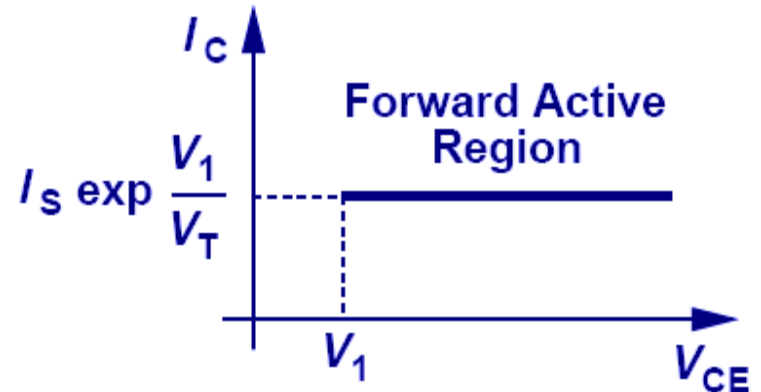
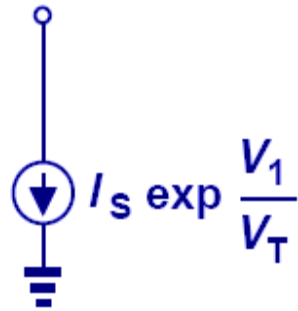
$$I_S = \frac{A_E q D_n n_i^2}{N_E W_B}$$

- Applying the law of diffusion, we can determine the charge flow across the base region into the collector.
- The equation above shows that the transistor is indeed a voltage-controlled element, thus a good candidate as an amplifier.

Constant Current Source



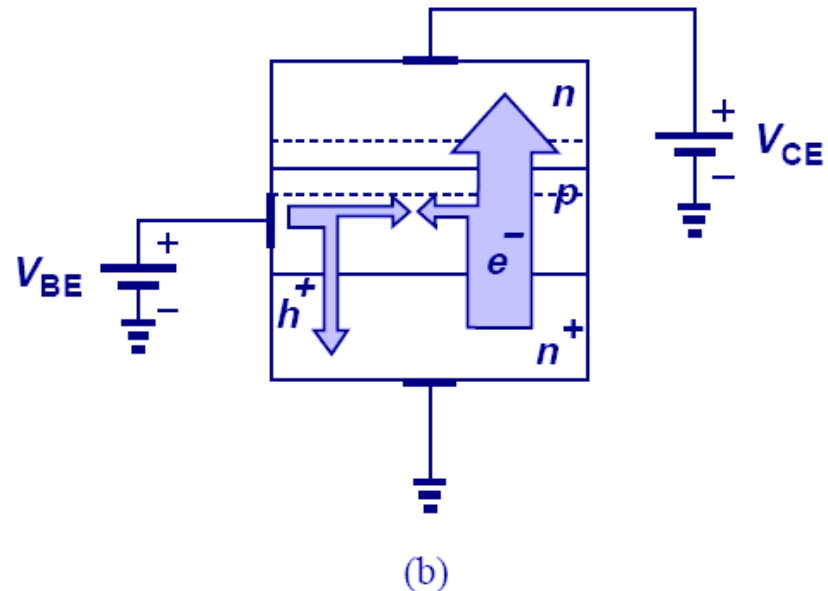
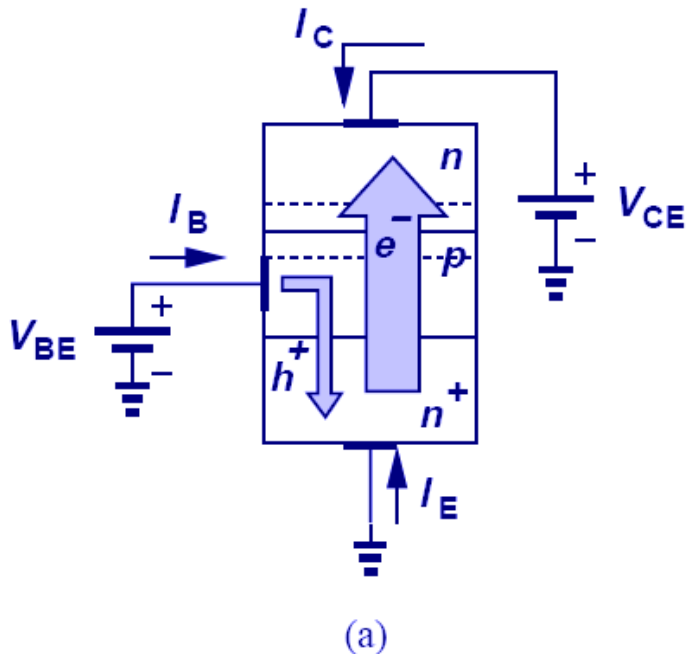
(a)



(b)

➤ **Ideally, the collector current does not depend on the collector to emitter voltage. This property allows the transistor to behave as a constant current source when its base-emitter voltage is fixed.**

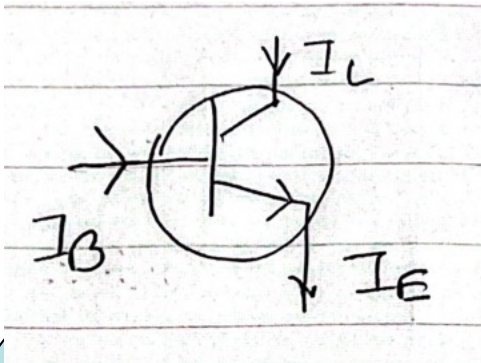
Base Current



$$I_C = \beta I_B$$

- **Base current consists of two components: 1) Reverse injection of holes into the emitter and 2) recombination of holes with electrons coming from the emitter.**

Emitter Current



$$I_E = I_C + I_B$$

$$I_E = I_C \left(1 + \frac{1}{\beta} \right)$$

$$\beta = \frac{I_C}{I_B}$$

➤ Applying Kirchoff's current law to the transistor, we can easily find the emitter current.

Summary of Currents

$$I_C = I_S \exp \frac{V_{BE}}{V_T}$$

$$I_B = \frac{1}{\beta} I_S \exp \frac{V_{BE}}{V_T}$$

$$I_E = \frac{\beta + 1}{\beta} I_S \exp \frac{V_{BE}}{V_T}$$

$$\frac{\beta}{\beta + 1} = \alpha$$

Review

- Section 4.2, 4.3
- 4.3.2
- Example 4.5,
Example 4.3